

Steps to download Hadoop on Ubuntu Operating system

Before starting the Hadoop installation: -

1. Make sure the system is connected to the Internet, because we need to install packages in order to successfully install Hadoop
2. Make sure the Ubuntu is updated by using the following command in the Ubuntu Terminal: -

sudo apt-get update

Steps for Hadoop installation: -

1. Install Java-jdk using: -

sudo apt-get install default-jdk

- a. Enter “Y” when prompted to proceed with the installation
- b. Wait for the installation to complete (installation may take time depending on speed of internet)
- c. To verify the proper installation of java-jdk use: -

java --version

2. Now we create a new group and user: -

- a. Create the group using the following command: -

sudo addgroup hadoop

- b. Create a new user: -

- i. Use command : - ***sudo adduser --ingroup hadoop hduser***

(Note : - You can use any name in place of hduser, just remember it)

- ii. After entering the command

1. Enter the password you want to set and then retype to confirm and hit Enter
2. Don't write anything for Full name, Room number, Work Phone, Home Phone, other. Just Press Enter for each of the line.
3. Then press “Y” to confirm the prompt of “is information entered correct?”
4. And the new user will be created.

3. Install ssh: -

- a. Use : - ***sudo apt-get install ssh***

- b. Enter “Y” when prompted

4. To create and setup SSH in hduser

- a. Now to we want to install hadoop in the new user i.e. “hduser”, so for that we need to move from current user to the new user that we created: -

- i. Use command – ***su hduser***

- ii. Enter the password that was used when creating the new user

- iii. The move to the home directory of the hduser user using command – ***cd***

- b. Setup ssh

- i. Enter the command: - ***ssh-keygen***

- ii. Don't type anything, just press enter for: -

1. Enter file in which to save the key
2. Enter passphrase
3. Enter same passphrase again:

- c. Use the following commands to start the ssh service
 - i. ***cat \$HOME/.ssh/id_rsa.pub >> \$HOME/.ssh/authorized_keys***
 - ii. ***ssh localhost***

5. Download Hadoop

- a. Paste this link on your browser in Ubuntu
<https://archive.apache.org/dist/hadoop/common/hadoop-3.3.4/>
- b. And download 3.3.4.tar.gz file
- c. By default the file will be downloaded in the Download folder in ubuntu.
- d. Copy the file from Downloads and paste it in the Home folder
- e. This file is been downloaded in the default ubuntu user and not the new user that we created
- f. To copy hadoop file from default user to new user (hduser)
 - i. Move to the default user using – ***su user_name*** (use the user name of your system)
 - ii. As our file to be moved is in the home folder, go to the home directory using ***cd*** command
 - iii. To move the file from default user to hduser, use the following command: -
sudo cp hadoop-3.3.4.tar.gz /home/hduser
- g. To verify the file is moved
 - i. Go back to the hduser using – ***su***
 - ii. Move to the home directory using – ***cd***
 - iii. To check the contents of the file use – ***ls***
- h. Before extracting the tar file we need to make the “hduser” a root user, so as to be able to use the “sudo” commands, to do so: -
 - i. Go to the ubuntu default user using – ***su user_name*** (use the name of your system)
 - ii. Go to the home directory using – ***cd***
 - iii. So make hduser a root user, use: - ***sudo adduser hduser sudo***
 - iv. Then move back to the hduser using – ***su***
 - v. To move to the home directory use – ***cd***
- i. Now that hduser is root user, we can extract the hadoop tar file
 - i. Use the command: - ***sudo tar xvf hadoop-3.3.4.tar.gz***
 - ii. After pressing enter for the above command extraction process will start
 - iii. After completion of the extraction process, we can confirm it by using - ***ls***
 - iv. Then create a new folder using - ***sudo mkdir -p /usr/local/hadoop***
 - v. Move inside the hadoop files extracted folder using - ***cd hadoop-3.3.4***
 - vi. Then move all the extracted files of hadoop in the newly created folder using:
sudo mv */usr/local/hadoop
 - vii. Move to the home folder using – ***cd***
 - viii. Then use: - ***sudo chown -R hduser:hadoop /usr/local/hadoop***
- j. Now we need to do some configurations
 - i. For that we need the path of the location where java-jdk is installed
 - 1. Open a new terminal and use - ***update-alternatives --config java***
 (Note: - Do not close the old terminal)
 - ii. Go back to the old terminal and enter the following command : -
sudo nano ~/.bashrc

- iii. Scroll down to the very bottom and paste this

```
#HADOOP VARIABLES START
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-i386
export HADOOP_INSTALL=/usr/local/hadoop
export PATH=$PATH:$HADOOP_INSTALL/bin
export PATH=$PATH:$HADOOP_INSTALL/sbin
export HADOOP_MAPRED_HOME=$HADOOP_INSTALL
export HADOOP_COMMON_HOME=$HADOOP_INSTALL
export HADOOP_HDFS_HOME=$HADOOP_INSTALL
export YARN_HOME=$HADOOP_INSTALL
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_INSTALL/lib/native
export HADOOP_OPTS="-Djava.library.path=$HADOOP_INSTALL/lib"
#HADOOP VARIABLES END
```

- iv. Now we need to change the path of the **JAVA_HOME**

1. From the other terminal where we opened the path for java, copy the path till amd64
2. And paste it in the JAVA_HOME line of of ~/.bashrc

- v. To save the changes enter : - **Ctrl + o**

- vi. Then press **enter**

- vii. To exit the nano file press : - **Ctrl + x**

- viii. To run the file for the newly made changes use : - **source ~/.bashrc**

- k. Now configuring another file: -

- i. Enter the command: -

sudo nano /usr/local/hadoop/etc/hadoop/hadoop-env.sh

- ii. And then paste this line: -

export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64

- iii. To save the changes enter: - **Ctrl + o**

- iv. Then press **enter**

- v. To exit the nano file press: - **Ctrl + x**

- l. Now enter the following commands: -

i. sudo mkdir -p /app/hadoop/tmp

ii. sudo chown hduser:hadoop /app/hadoop/tmp

m. Now we need to make changes in the core-site.xml file

i. Use the following command to get inside the core-site.xml file

sudo nano /usr/local/hadoop/etc/hadoop/core-site.xml

ii. Paste the following lines between <configuration> and </configuration>

```
<property>
<name>hadoop.tmp.dir</name>
<value>/app/hadoop/tmp</value>
<description>A base for other temporary
directories.</description>
</property>
<property>
<name>fs.default.name</name>
<value>hdfs://localhost:54310</value>
<description>The name of the default file system.
A URI whose
scheme and authority determine the FileSystem implementation.
The
uri's scheme determines the config property (fs.SCHEME.impl)
naming
the FileSystem implementation class.
used to
The uri's authority is
determine the host, port, etc. for a filesystem.</description>
</property>
```

iii. To save the changes enter: - **Ctrl + o**

iv. Then press **enter**

v. To exit the nano file press: - **Ctrl + x**

n. Similarly we need to configure mapred-site.xml

i. Open the file using :-

sudo nano /usr/local/hadoop/etc/hadoop/mapred-site.xml

ii. Between <configuration> and </configuration> paste: -

```
<property>
<name>mapred.job.tracker</name>
<value>localhost:54311</value>
<description>The host and port that the MapReduce job tracker
runs
at.
If "local", then jobs are run in-process as a single map
and reduce task.
</description>
</property>
```

iii. To save the changes enter: - **Ctrl + o**

iv. Then press **enter**

v. To exit the nano file press: - **Ctrl + x**

o. Create folder for namenode and datanode

i. ***sudo mkdir -p /usr/local/hadoop_store/hdfs/namenode***

ii. ***sudo mkdir -p /usr/local/hadoop_store/hdfs/datanode***

iii. ***sudo chown -R hduser:hadoop /usr/local/hadoop_store***

- p. Configure hdfs-site.xml file
 - i. Open the file using: -
sudo nano /usr/local/hadoop/etc/hadoop/hdfs-site.xml
 - ii. Enter these lines between <configuration> and </configuration>

```
<property>
<name>dfs.replication</name>
<value>1</value>
<description>Default block replication.
The actual number of replications can be specified when the
file is created.
The default is used if replication is not specified in create
time.
</description>
</property>
<property>
<name>dfs.block.size</name>
<value>1048576</value>
</property>
<property>
<name>dfs.namenode.name.dir</name>
<value>file:/usr/local/hadoop_store/hdfs/namenode</value>
</property>
<property>
<name>dfs.datanode.data.dir</name>
<value>file:/usr/local/hadoop_store/hdfs/datanode</value>
</property>
```

- iii. To save the changes enter: - **Ctrl + o**
 - iv. Then press **enter**
 - v. To exit the nano file press: - **Ctrl + x**
- 6. Now all files are configured
- 7. To use the hadoop system first we need to run the following command: -
hadoop namenode -format
- 8. Then to start the hdfs and yarn system we use: -
 - a. ***start-dfs.sh***
 - b. ***start-yarn.sh***
- 9. To verify all services are working use: - ***jps***
- 10. We can also confirm by typing : localhost:9870 in ubuntu browser to check if data node is working properly
- 11. Similarly to confirm resource manager working properly we can go to : localhost:8088
- 12. If it works properly Hadoop Installation is now complete